UECS3294 ADVANCED WEB APPLICATION DEVELOPMENT

ASSIGNMENT

January 2026 Trimester

Course Outcomes

In this assessment, you are assessed based on the following **course outcomes**:

CO1	Build interactive and database-driven web applications using a web application frame- work and model-view-controller (MVC) software architecture.
CO2	Build web applications with user authentication and authorization using cookies, session and role-based access control (RBAC).

Assessment Contribution

This assignment contributes **20%** to the overall assessment for this course.

Membership

This is a **group** assignment: You should have **FOUR (4)** students per group.

Deadline

The deadline for this assignment is the end of Week 12.

1. Task

Build a web application based on the ***Laravel framework*** with **a topic of your choice**. This application should conform to the following requirements:

1. Must follow Laravel's **MVC structure properly.**
2. Must have a **MySQL** database.
3. All the operations should be done using the **model (Eloquent ORM)** approach.
4. The database should include users table plus at least **TWO** more tables (excluding pivot tables) with a relationship.
5. The **CRUD** operation should be done with data validation.
6. Must implement **user authentication**.
7. Must demonstrate the use of **sessions or cookies**.
8. Must Implement User **authorization** using **gates** or **policies**.

Registration of Title

Please register your chosen group members and title in the Google sheet that have shared through WBLE. *Ensure that you are signed in to your UTAR Account (**@1utar.my**) before accessing the registration Google Sheet*. **It is recommended to not more groups having similar web application titles should be registered into the list.

2. Deliverables

You should submit the following deliverables.

A **ZIP** archive named using your group number (e.g. **UECS3294_Assign_GroupNumber.zip**) and include the following:

1) Archived softcopy of your Laravel project, including your database SQL file. **Please remove the vendor folder from your project zip file during the uploading to minimize the file size (compulsory).
2) PDF document/report (25 pages) containing the following sections:
   Title and **description** of your web application explaining the samples of codes as follows:
   1. Database design and migrations
   2. Models and relationships
   3. CRUD Operations
   4. Input validation
   5. Relational queries and search filters
   6. Authentication logic
   7. Authorization logic
   8. Cookies and session implementation.
   9. Include the **marking rubric page at the end of your report** as the final section.
3) Submit a **15-minute recorded demo video** showing your project features and brief code walkthrough (full video requirements will be shared in a separate document).

Submission & Originality Policies

1. This assignment must be submitted completely by the abovementioned deadline.
2. All work must be original and if, taken from any works other than yours must be properly referenced.
3. Any plagiarized content will be penalized, including the award of zero marks.